Cartan path-space exactness correction and the fixed-skew Sobolev

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

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1. Purpose and scope

R-119 and R-120 left a rational first-order Cartan term and proposed a checksum: the isolated R-102 current has repository curl $-40/729$, so the unwritten remainder of the complete terminal owner was said to be required to provide $+40/729$. The present note audits that implication before spending more proof effort on an arithmetic companion search.

The audit finds a category error. Exactness of the differential of a scalar functional on path or jet space does not imply that a target-space one-form appearing as one local coefficient is closed. The complete rational owner does telescope exactly, but its recombined local current is itself not closed. There is therefore no structurally mandatory $+40/729$ local companion. The observed $-40/729$ and its chain-primitive no-go remain valid.

The correction unlocks a different analytic route. A fixed-skew Cartan form paired with an H^{-s} coefficient has positive deterministic Young slack for every $0 \leq s < 1$. At $s = 3/5$ it requires only a fifth coefficient moment, improving the older R-071 Cartan payment on any branch already reduced to its unshifted stationary one-form current. A sharp high-frequency fixture proves that the R-120 zeroth-order $H^{-11/10}$ coefficient class cannot be reused for this first-order form.

> **R-121 conclusion.** At fixed cutoff, positive floor, and one common legal target heat, the two-visit rational owner $\mathcal{R}_Q + \mathcal{M}_U + \mathcal{K}_R$ telescopes pathwise to its two terminal endpoints. Target heat is internal, the covariance trace occurs once, and the intermediate raw-Wick/forest endpoint cancels after complete recombination.
> On the normalized R-102 slice, the repository curls of the $\mathcal{K}_R$ current, the $\mathcal{M}_U$ current, and their recombined current are respectively $-40/729$, $2720/729$, and $2680/729$. Thus neither the observed owner nor global scalar exactness supplies the alleged $+40/729$ local companion.
> For fixed skew matrices and $0 \leq s < 1$, the surviving Cartan form obeys a one-use estimate with coefficient moment $2/(1-s)$. At $s = 3/5$ the powers are $X^{7/10}Y^{1/10}$, the slack is $1/5$, and the required moment is 5. The adapted production current is not yet reconstructed in that class.
> Adapted D_0, D_1, an adapted R-063 forest theorem or equivalent fifth $H^{-3/5}$ moment, the complete one-use aggregation, $\mathrm{OVERLAP}_{\mathrm{src}}$, Nelson, removals, interacting measure, and Sector-A closure remain open. Tier stays T4.

R-121 supersedes only the local-companion inference in R-119 Section 6, R-120 Section 9, and the matching parts of EXP-000370 and EXP-000375. The historical notes and certificates remain immutable evidence of what was tested. All other R-119/R-120 results survive this correction.

2. Exact two-visit rational owner

Fix one spatial derivative index and a legal R-119 two-block/two-visit chart. For $k = 1, 2$, write

$$U_k = U_{k-1} + a_k, \qquad G_k = G_{k-1} + d_k, \qquad d_k = D_i a_k, \tag{2.1}$$

where $a_2 = S_2 h_2(\xi_1)$ is strict-past for the second block. At one common deterministic or legally predictable target heat and covariance, put

$$B_k = (P_{\Sigma_{\rm tar}} B_R)(U_k), \qquad Q_k = G_k^{\otimes 2} - \Gamma_i. \tag{2.2}$$

Define the left Taylor coefficient and its exact remainder by

$$T_k = B_{k-1} + DB_{k-1}[a_k] + \frac{1}{2} D^2 B_{k-1}[a_k, a_k], \qquad L_k = B_k - T_k. \tag{2.3}$$

The three rational owner coordinates are

$$\mathcal{R}_{Q,k} = \frac{1}{2}(B_k - B_{k-1}) : Q_{k-1}, \tag{2.4}$$

$$\mathcal{M}_{U,k} = G_{k-1}^T T_k d_k, \tag{2.5}$$

$$\mathcal{K}_{R,k} = G_{k-1}^T L_k d_k + \frac{1}{2} d_k^T B_k d_k. \tag{2.6}$$

Theorem 2.1 (exact two-visit owner telescope)

Pathwise at every finite cutoff,

$$\boxed{\sum_{k=1}^{2} (\mathcal{R}_{Q,k} + \mathcal{M}_{U,k} + \mathcal{K}_{R,k}) = \frac{1}{2} B_2 : Q_2 - \frac{1}{2} B_0 : Q_0.} \tag{2.7}$$

Equivalently,

$$\boxed{\frac{1}{2}(G_2^T B_2 G_2 - G_0^T B_0 G_0) - \frac{1}{2}(B_2 - B_0) : \Gamma_i.} \tag{2.8}$$

Proof. Since $G_k = G_{k-1} + d_k$,

$$Q_k = Q_{k-1} + G_{k-1} \otimes d_k + d_k \otimes G_{k-1} + d_k \otimes d_k. \tag{2.9}$$

Therefore

$$\begin{aligned} \frac{1}{2} B_k : Q_k - \frac{1}{2} B_{k-1} : Q_{k-1} = \frac{1}{2}(B_k - B_{k-1}) : Q_{k-1} \\ + G_{k-1}^T B_k d_k + \frac{1}{2} d_k^T B_k d_k. \end{aligned} \tag{2.10}$$

Using $B_k = T_k + L_k$, the right side is exactly (2.4)–(2.6). Summing (2.10) cancels $B_1 : Q_1/2$ and proves (2.7). No independence or conditioning is used. Thus adapted dependence of a_2 does not affect the finite-cutoff algebra; it matters only when conditional expectation, Wiener projection, or continuum reconstruction is subsequently taken. □

Proposition 2.2 (once-only owner incidence)

The terminal derivative square is endpoint two minus endpoint zero, once. The covariance-normal trace is $-(B_2 - B_0) : \Gamma_i/2$, once. Target heat is already internal to $P_{\Sigma_{\rm tar}} B_R$ and is not appended as another owner. The two surviving endpoint raw-Wick objects are reconstructed once; the intermediate endpoint has multiplicity zero after recombination.

Consequently the R-063 P_3/P_1 and $P_4/P_2^{\rm cross}/\Sigma Q/P_0^{\rm cross}$ families, together with the Taylor $\mathcal{R}_3$ family, are coordinates of the two endpoint reconstructions, not repairs that may be appended per visit. The accepted R-063 continuum theorem remains deterministic-shift only. Equation (2.7) does not silently promote it to adapted $h_2(\xi_1)$.

The covariance-normal trace in (2.8) and the R-119 root-output trace $\delta\tau$ are also distinct objects until a coefficient-level identity relates them. This distinction is load-bearing for D_0, D_1 below.

3. Exact normalized owner-current audit

Use the R-102 active/kernel slice with repository curl convention

$$\mathrm{curl}_{\mathrm{repo}}(v) = \partial_y v_x - \partial_x v_y. \tag{3.1}$$

Set $\alpha = 5/9$, $d = 1 + x^2 + y^2$, and

$$g(x,y) = \left(x - \alpha\frac{x^3}{d}, -\alpha\frac{x^2 y}{d}\right), \qquad B = 4gg^T. \tag{3.2}$$

For $a = c = e_1$, define

$$B_0 = B(x,y), \quad B_1 = B(x+1,y), \quad B_T = B_0 + \partial_x B_0 + \frac{1}{2}\partial_x^2 B_0, \quad L = B_1 - B_T. \tag{3.3}$$

At $(x,y) = (1,1)$, exact rational differentiation gives

$$B_0 = \frac{1}{729}\begin{pmatrix} 1936 & -440 \\ -440 & 100 \end{pmatrix}, \quad B_1 = \frac{1}{729}\begin{pmatrix} 4624 & -1360 \\ -1360 & 400 \end{pmatrix}, \tag{3.4}$$

$$L = \frac{1}{2187}\begin{pmatrix} 80 & 520 \\ 520 & -300 \end{pmatrix}. \tag{3.5}$$

Proposition 3.1 (three owner-current curls)

The current coefficients and curls are

$$\omega_K = Le_1 = \frac{1}{2187}(80, 520)^T, \qquad \mathrm{curl}_{\mathrm{repo}}\,\omega_K = -\frac{40}{729}, \tag{3.6}$$

$$\omega_M = B_T e_1 = \frac{1}{2187}(13792, -4600)^T, \qquad \mathrm{curl}_{\mathrm{repo}}\,\omega_M = \frac{2720}{729}, \tag{3.7}$$

$$\omega_{M+K} = B_1 e_1 = \frac{1}{729}(4624, -1360)^T, \qquad \mathrm{curl}_{\mathrm{repo}}\,\omega_{M+K} = \frac{2680}{729}. \tag{3.8}$$

More explicitly,

$$\partial_y(\omega_K)_x = -\frac{4480}{6561}, \qquad \partial_x(\omega_K)_y = -\frac{4120}{6561}, \tag{3.9}$$

$$\partial_y(\omega_M)_x = \frac{20800}{6561}, \qquad \partial_x(\omega_M)_y = -\frac{3680}{6561}. \tag{3.10}$$

Thus ω_M is not an arithmetic $+40/729$ companion, and even the recombined coefficient $\omega_{M+K} = \omega_M + \omega_K$ is not closed. The standard $dx \wedge dy$ orientation is the negative of (3.1), so the isolated standard exterior coefficient is $+40/729$; this convention change does not alter the conclusion.

The R-102 $-40/729$ observation still proves that the isolated Le_1 cannot be a target-space chain primitive. What fails is only the additional claim that another local owner must carry its arithmetic opposite.

4. Path-space exactness does not imply target-space closure

Let θ be a smooth one-form on a finite-dimensional target and let u be a smooth loop. Define the scalar first-order functional

$$I[u] = \int_{\mathbb{T}} \theta_{u(t)}(\partial_t u(t))\,\mathrm{d}t. \tag{4.1}$$

Theorem 4.1 (path-space exactness lemma)

The differential DI is an exact one-form on loop space by construction, even when $d\theta \neq 0$ on the target. For fixed variations z, w,

$$DI[u]z = \int_{\mathbb{T}} (d\theta)_u(z, \partial_t u)\,\mathrm{d}t, \tag{4.2}$$

$$D^2 I[u](z, w) = \int_{\mathbb{T}} \left[(D_w d\theta)_u(z, \partial_t u) + (d\theta)_u(z, \partial_t w)\right]\,\mathrm{d}t. \tag{4.3}$$

Although (4.3) is not manifestly symmetric, the difference under $z \leftrightarrow w$ vanishes by $d(d\theta) = 0$ and integration by parts on the torus. Hence symmetry of the scalar Hessian does not force $d\theta = 0$.

Explicit falsifying example. Take

$$\theta = -\frac{y}{2}\,\mathrm{d}x + \frac{x}{2}\,\mathrm{d}y, \qquad d\theta = \,\mathrm{d}x \wedge \,\mathrm{d}y, \tag{4.4}$$

and $u(t) = (\cos t, \sin t)$. Then $I[u] = \pi$. For $z = (\cos t, 0)$ and $w = (0, \sin t)$, direct integration gives

$$D^2 I[u](z, w) = D^2 I[u](w, z) = \pi. \tag{4.5}$$

Thus a nonclosed target one-form produces a perfectly exact path-space differential and a symmetric scalar Hessian.

There is also a production-specific version. Let $z_0 = (1, 1)$, $\omega = Le_1$, use normalized Haar measure on the loop parameter, and set

$$u_{\varepsilon,\delta}(t) = z_0 + \varepsilon e_1 \cos t + \delta e_2 \sin t. \tag{4.6}$$

For $I(\varepsilon, \delta) = \langle \omega(u) \cdot \partial_t u \rangle$, the affine jet of ω at z_0 gives exactly

$$I_{\mathrm{aff}}(\varepsilon, \delta) = \frac{1}{2}\left(\partial_x \omega_y - \partial_y \omega_x\right) \varepsilon\delta = \frac{20}{729}\varepsilon\delta. \tag{4.7}$$

Hence both mixed derivative orders equal $20/729$, while the repository curl is $-40/729$. This directly falsifies the old implication on the actual R-102 jet, not merely on the illustrative form (4.4).

Apply this to the endpoint scalar

$$\mathcal{E}(U, G) = \frac{1}{2} B(U) : (G^{\otimes 2} - \Gamma). \tag{4.8}$$

The complete owner is the scalar difference $\mathcal{E}(U_2, G_2) - \mathcal{E}(U_0, G_0)$. Extracting one coefficient linear in G and then taking the target exterior derivative is not a chain map from the full (U, G) differential. Equations (3.6)–(3.8) are an exact production-specific witness. This proves the R-121 correction.

5. Fixed-skew first-order Sobolev theorem

Let $A_\nu^* = -A_\nu$ be a fixed finite family of real matrices. Put

$$\mathbf{F}(z)_{\nu i} = z^T A_\nu \partial_i z, \qquad \mathcal{C}_Q(z) = \sum_{\nu, i} \langle Q_{\nu i}, \mathbf{F}(z)_{\nu i} \rangle. \tag{5.1}$$

The coefficient norm is the vector-valued norm

$$M_s = \|\mathbf{Q}\|_{H^{-s}(\ell^2)}, \qquad 0 \leq s < 1. \tag{5.2}$$

Let C_{sk} denote the fixed-family constant in the vector-valued pointwise skew estimate. For the canonical six-real basis

$$A_{pq} = \frac{1}{2}(E_{pq} - E_{qp}), \qquad 1 \leq p < q \leq 6, \tag{5.3}$$

the exact wedge identity is

$$\sum_{p<q} |z^T A_{pq} v|^2 = \frac{1}{4}\left(|z|^2|v|^2 - (z \cdot v)^2\right), \tag{5.4}$$

so one may take the algebraic ℓ^2 constant $C_{\mathrm{sk}} = 1/2$ before the universal Sobolev product constant. Separately,

$$\sum_{p<q} |A_{pq}| = \frac{5}{2}\mathrm{I}_6. \tag{5.5}$$

Equation (5.5) is an absolute-operator inventory, not the coefficient-index ℓ^2 constant in (5.4); conflating them would produce a norm-convention error.

Write

$$X = \|z\|_{H^2}^2, \qquad Y = \|z\|_6^6. \tag{5.6}$$

Lemma 5.1 (two endpoint product estimates)

In three dimensions,

$$\|\mathbf{F}(z)\|_{L^2(\ell^2)} \le C C_{\mathrm{sk}} \|z\|_6 \|\nabla z\|_3 \le C C_{\mathrm{sk}} X^{1/4} Y^{1/4}, \tag{5.7}$$

and

$$\|\mathbf{F}(z)\|_{H^1(\ell^2)} \le C C_{\mathrm{sk}} X. \tag{5.8}$$

Proof. Equation (5.7) uses (5.4), Holder, and the three-dimensional Gagliardo–Nirenberg inequality $\|\nabla z\|_3 \le C\|D^2 z\|_2^{1/2}\|z\|_6^{1/2}$, with harmless lower Fourier modes absorbed in H^2. Differentiating (5.1) gives products of ∇z with itself and of z with $D^2 z$. The embeddings $H^2 \hookrightarrow L^\infty$ and $H^2 \hookrightarrow W^{1,4}$ prove (5.8). □

Theorem 5.2 (sharp fixed-skew Sobolev one-use boundary)

Complex interpolation of (5.7)–(5.8) gives

$$\boxed{\|\mathbf{F}(z)\|_{H^s(\ell^2)} \le C_s C_{\mathrm{sk}} X^{(1+3s)/4} Y^{(1-s)/4}, \qquad 0 \le s < 1.} \tag{5.9}$$

The two deterministic exponents sum to $(1+s)/2$, leaving Young slack $(1-s)/2$. Vector-valued duality and weighted Young therefore give, for every $\eta, \zeta > 0$,

$$\boxed{|\mathcal{C}_Q(z)| \le \eta X + \zeta Y + C_s C_{\mathrm{sk}}^{2/(1-s)} \eta^{-(1+3s)/(2(1-s))} \zeta^{-1/2} M_s^{2/(1-s)}.} \tag{5.10}$$

At $s = 3/5$,

$$\boxed{|\mathcal{C}_Q(z)| \le \eta X + \zeta Y + C C_{\mathrm{sk}}^5 \eta^{-7/2} \zeta^{-1/2} M_{3/5}^5.} \tag{5.11}$$

The exact ledger is

$$(X\text{-power}, Y\text{-power}, \text{slack}) = \left(\frac{7}{10}, \frac{1}{10}, \frac{1}{5}\right), \tag{5.12}$$

and the coefficient moment is five.

Corollary 5.3 (covariance-horizontal form)

Let

$$A_0 = \frac{M_R^2}{c_{\mathrm{sym}}}, \quad S = \|h\|_{\mathcal{S}}^2, \quad T = \|Z_h\|_6^6, \quad B = \|X_J\|_6^6. \tag{5.13}$$

R-120 gives $X \le A_0 S$ and $Y \le 32(T+B)$. Choosing $\eta = \varepsilon/A_0$ and $\zeta = \varepsilon_6/32$ in (5.11) yields

$$\boxed{|\mathcal{C}_Q(z)| \le \varepsilon S + \varepsilon_6(T+B) + C\sqrt{32}\, C_{\mathrm{sk}}^5 A_0^{7/2} \varepsilon^{-7/2} \varepsilon_6^{-1/2} M_{3/5}^5.} \tag{5.14}$$

Thus a cutoff-, chart-, visit-, and control-uniform fifth moment of the correct reconstructed $H^{-3/5}$ Cartan coefficient would close this component with one source/sextic payment.

6. What R-071 already pays and what it does not

R-071 proves, for an unshifted common real-even stationary Gaussian field U and every finite p,

$$\Theta(U)\partial_i U \in L^p(\Omega; H^{-1/2-\delta}). \tag{6.1}$$

Taking $\delta = 1/10$ gives the fifth $H^{-3/5}$ moment required by (5.11). Therefore every R-120 Cartan branch that is algebraically reduced to a finite sum of unshifted R-071 currents paired with fixed skew matrices is now directly payable. The deterministic interpolation sharpens the old R-071 45/4 Cartan moment arithmetic to moment 5 on this fixed-skew form.

This is not yet the adapted production theorem. The raw coefficient there is evaluated along $u = X + tz$ and has the schematic form

$$\widetilde{\Theta}(X+tz)\partial_i(X+tz). \tag{6.2}$$

R-063 proves its finite forest for deterministic translation; R-075 records that arbitrary adapted substitution need not preserve that finite forest. Neither authority supplies a control-uniform fifth $H^{-3/5}$ moment for (6.2). The minimal honest additional input is an owner-complete adapted Cartan Taylor/forest identity reducing (6.2) to the unshifted R-071 current plus explicitly controlled translated corrections, or an equivalent direct uniform fifth-moment theorem.

7. Sharp failure of the rough first-order reuse route

Work on a normalized one-dimensional torus embedded in $\mathbb{T}^3$, let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad z_N = (1, N^{-2}\sin Nx), \qquad Q_N = N^s \cos Nx. \tag{7.1}$$

Then $\|z_N\|_{H^2}$ and $\|z_N\|_6$ are bounded uniformly in N, while

$$\|Q_N\|_{H^{-s}}^2 = \frac{N^{2s}}{2(1+N^2)^s} \le \frac{1}{2}. \tag{7.2}$$

But

$$\langle Q_N, z_N^T J \partial_x z_N \rangle = -\frac{1}{2} N^{s-1}. \tag{7.3}$$

For $s = 11/10$, (7.3) diverges as $N^{1/10}$. Hence an absolute first-order Cartan bound from only H^2, L^6, and $H^{-11/10}$ norms is impossible. At $s = 1$ the positive Young slack vanishes; for $s < 1$ it is exactly $(1-s)/2$. This proves that the R-120 zeroth-order coefficient class and the present first-order class have different sharp Sobolev boundaries.

8. Adapted low-chaos coefficients remain independent

Let the complete root-one output be

$$Y = b + A\xi_1 + R, \qquad R = \sum_{n\geq 2} I_n(r_n), \qquad \delta\tau = \sum_{n\geq 0} I_n(t_n). \tag{8.1}$$

The exact R-119 reductions are

$$\boxed{D_0 = t_0 - \sum_{n\geq 2} n!\|r_n\|^2,} \tag{8.2}$$

$$\boxed{(D_1)_i = (t_1)_i - 4\sum_j \langle Ae_j, r_2(i,j)\rangle - 2\sum_{n\geq 3} n!\langle r_{n-1}, r_n(\cdot, i)\rangle.} \tag{8.3}$$

For the adapted endpoint,

$$V = D_{\xi_1} Z_* = S_1 + S_2 Dh_2, \qquad W = D^2_{\xi_1} Z_* = S_2 D^2 h_2, \tag{8.4}$$

so

$$D^2_{\xi_1} B(Z_*) = D^2 B(Z_*)[V,V] + DB(Z_*)[W] \tag{8.5}$$

contains the four independent families

$$D^2B[S_1, S_1], \quad 2D^2B[S_1, S_2Dh_2], \quad D^2B[S_2Dh_2, S_2Dh_2], \quad DB[S_2D^2h_2]. \tag{8.6}$$

The current authorities do not specify the needed adapted kernels of h_2 or the corresponding trace/forest kernels t_0, t_1. Thus (8.2)–(8.3) cannot yet be evaluated.

The independence is falsifiable in the scalar diagnostic

$$R = \alpha H_2 + \beta H_3, \qquad \delta\tau = q_0 + q_1 G. \tag{8.7}$$

Exact Gaussian moments give

$$D_0 = q_0 - (2\alpha^2 + 6\beta^2), \qquad D_1 = q_1 - (4a\alpha + 12\alpha\beta). \tag{8.8}$$

The owner telescope in Section 2 holds for arbitrary q_0, q_1, so it cannot force either expression in (8.8) to vanish. This is a method counterexample, not an A1 production counterexample.

9. Devil's-advocate review

1. **Objection:** scalar terminal exactness makes every extracted current coefficient closed. **Disposition: UPHELD AGAINST THE OLD INFERENCE.** Theorem 4.1 and (3.8) give an explicit counterexample.

2. **Objection:** $\mathcal{M}_U$ is the missing $+40/729$ companion. **Disposition: DISMISSED.** Its exact repository curl is $2720/729$.

3. **Objection:** $5/2$ is automatically the sharp coefficient-index constant for the canonical skew basis. **Disposition: VALID WITH CORRECTION.** It is the absolute-operator sum. The canonical ℓ^2 wedge constant is $1/2$; the theorem states its norm convention explicitly.

4. **Objection:** the R-120 $H^{-11/10}$ model class can be inserted unchanged into the first-order Cartan form. **Disposition: DISMISSED.** The exact family (7.1) diverges as $N^{1/10}$.

5. **Objection:** R-071 already proves the adapted coefficient moment. **Disposition: VALID BOUNDARY.** R-071 is unshifted stationary; R-063 and R-075 prevent an automatic adapted finite-forest transfer.

6. **Objection:** the pathwise owner telescope proves $D_0 = D_1 = 0$. **Disposition: DISMISSED.** Equation (8.8) varies with independent trace data while the telescope remains unchanged.

10. Corrected proof route and exact remaining gate

The arithmetic companion search is retired. The shortest noncyclic route is now parallel but sharply separated:

1. reconstruct the complete adapted endpoint Wiener kernels entering t_0, t_1, r_2 and the adjacent-chaos contractions in (8.2)–(8.3);
2. derive an owner-complete Cartan Taylor/forest identity that places the surviving coefficient in $L^5(\Omega; H^{-3/5})$, or prove the equivalent uniform fifth-moment statement directly;
3. apply (5.14) once over the R-120 covariance-horizontal synthesis and then assemble the remaining R-118 zeroth-order owner without duplicating source energy or terminal sextic budgets.

An exact Euler–Lagrange cancellation of the complete endpoint scalar remains possible, but it must be derived before projection into current, trace, and forest coordinates. No local opposite curl may be inserted as a checksum.

11. Result footer

The primary exact symbolic executable independently derives the three current curls, the generic symmetric two-visit identity, the path-space example, the canonical skew identities, every Sobolev/Young exponent, the high-frequency growth exponent, and the scalar D_0, D_1 diagnostic. A non-importing standard-library executable uses exact bivariate Fraction jets, three rational two-visit fixtures, independent Gaussian polynomial moments, and independent skew-basis arithmetic. The integrated verifier reruns both, pins all authorities and source hashes, checks the paired PDF, and enforces the correction and no-overclaim surfaces.

R-121 does not prove adapted D_0, D_1 cancellation, an adapted R-063 forest, a production fifth $H^{-3/5}$ moment, Cartan cancellation, the full one-use source/sextic inequality, $\mathrm{OVERLAP}_{\mathrm{src}}$, Nelson, removal of cutoff or floor, construction of an interacting measure, Sector-A closure, or any T5–T7 promotion.

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Result ID: A13-CLASSII-CARTAN-PATHSPACE-EXACTNESS-FIXED-SKEW-
  SOBOLEV-BOUNDARY (R-121).
Precise statement: Thm. 2.1; Props. 3.1-3.2; Thm. 4.1;
  Lem. 5.1 and Thm. 5.2; Prop. 6.1; Thm. 7.1; Sec. 8.
Scope: fixed finite cutoff and positive rational floor; deterministic
  fixed-skew theorem; adapted production moment remains open.
Dependencies: A1, R-063, R-071, R-075, R-102, R-119, R-120.
Evidence grade: ANALYTIC, EXACT, EXECUTED.
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/a13_classii_cartan_pathspace_exactness_fixed_skew_
  sobolev_boundary_verify.py
Expected output: R-121 PASS 125/125; primary 78/78; independent
  64/64; all hash, PDF, route, registry, and no-overclaim gates pass.
Falsification gate: any failed telescope, curl, path-Hessian, skew,
  Sobolev, high-frequency, authority, result, PDF, or surface assertion.
Tier before / after: T4 / T4.
No-overclaim statement: adapted D0/D1, adapted fifth H^{-3/5}
  moment/forest, one-use, Nelson, removals, measure, Sector A remain open.
Next required action: compute the actual adapted endpoint Wiener kernels
  and fifth H^{-3/5} current/forest moment, then spend the form once.
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